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CUET UG

Chemistry

Chemical Kinetics

Official Syllabus Unit III · Class 12 · CUET UG 2027 Cycle Notes

INSIDE
Concept Notes

INSIDE
Solved Examples

INSIDE
Practice MCQs + Solutions

Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official Syllabus Pointer (NTA CUET UG Baseline)	Skills & Competencies Tested
Rate of Reaction & Rate Law	Rate of a reaction (average and instantaneous), factors affecting rate: concentration, temperature, catalyst; order and molecularity of a reaction.	Formulating stoichiometric rate expressions, determining order from rate laws or experimental initial-rate data tables, and contrasting order with molecularity.
Integrated Rate Equations & Half-Life	Integrated rate equations and half-life for zero and first order reactions; pseudo-first-order reactions; gaseous phase first order decomposition.	Applying $[R] = [R]_0 - kt$ and $k = \left(\frac{2.303}{t}\right)\log\left(\frac{[R]_0}{[R]}\right)$, computing $t_{\frac{1}{2}}$, solving multi-step fractional decay $(t_{75\%}, t_{99.9\%})$, and analyzing partial pressures.
Temperature Dependence & Arrhenius Equation	Concept of collision theory (elementary ideas, no mathematical treatment), activation energy, Arrhenius equation, catalysis energetics.	Applying $\log\left(\frac{k_2}{k_1}\right) = \left(\frac{E_a}{2.303}R\right)\left[\frac{T_2 - T_1}{T_1 T_2}\right]$, reading slopes of $\log k$ vs $\frac{1}{T}$, and evaluating the effect of catalysts on $E_a, \Delta H$, and K_c.

EXAM PATTERN

CUET UG Exam Strategy: Unit III Chemical Kinetics regularly yields 4 to 5 questions in the CUET UG paper (+5, −1 mark scheme). The highest-yield concepts tested include: (1) reading reaction order from the physical units of k (s^{-1} for 1st order, mol L^{-1} s^{-1} for zero order, L mol^{-1} s^{-1} for 2nd order), (2) the independence of 1st order half-life on initial concentration ($t_{\frac{1}{2}} = \frac{0.693}{k}$), (3) multi-step relations like $t_{75\%} = 2 \times t_{50\%}$ and $t_{99.9\%} \approx 10 \times t_{50\%}$, and (4) the fact that a catalyst lowers E_a but leaves ΔH and equilibrium constant K_c completely unchanged.

Concept Module 1: Rate of Chemical Reaction

Chemical Kinetics investigates the speeds of chemical reactions, the factors governing their rates, and the molecular mechanisms by which they proceed.

- **Thermodynamics vs Kinetics:** Thermodynamics predicts whether a reaction is feasible ($\Delta G < 0$); kinetics determines **how fast** that reaction occurs.
- **Classification by Speed:**
  - **Instantaneous Reactions:** Ionic reactions like precipitation of $AgCl$ ($AgNO_3 + NaCl \rightarrow AgCl\downarrow + NaNO_3$) occur within 10^{-12} to 10^{-16} seconds.
  - **Extremely Slow Reactions:** Rusting of iron, weathering of rocks require months or years.
  - **Moderate (Measurable) Reactions:** Molecular reactions like inversion of cane sugar, hydrolysis of starch, or decomposition of N_2O_5 proceed at measurable rates under ambient conditions.

Average Rate vs Instantaneous Rate

Consider a general reaction: $R \rightarrow P$

1. **Average Rate (r_{av}):** The change in molar concentration of reactants or products divided by the finite time interval (Δt):

$$r_{\text{av}} = -\frac{\Delta[R]}{\Delta t} = +\frac{\Delta[P]}{\Delta t}$$

   The negative sign denotes that reactant concentration **decreases** over time.

2. **Instantaneous Rate (r_{inst}):** The rate of reaction at a specific infinitesimal instant of time ($\Delta t \rightarrow 0$), determined from the slope of the tangent to the concentration-time curve:

$$r_{\text{inst}} = -\frac{d[R]}{dt} = +\frac{d[P]}{dt}$$

Stoichiometric Rate Expressions

For a reaction with arbitrary stoichiometric coefficients:

$$aA + bB \rightarrow cC + dD$$

The rate of reaction is defined by dividing the individual rates of disappearance or appearance by their stoichiometric coefficients:

$$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt}$$

- **Units of Reaction Rate:** Always mol L^{-1} s^{-1} (or mol L^{-1} min^{-1}). For gaseous reactions: atm s^{-1} or bar s^{-1}.

WORKED EXAMPLE 1 — STOICHIOMETRIC RATE RELATIONS

In the decomposition of dinitrogen pentoxide:

$$2N_2O_5(g) \rightarrow 4NO_2(g) + O_2(g)$$

If the rate of formation of NO_2 is 0.0072 mol L^{-1} s^{-1}, calculate: (a) the rate of reaction, and (b) the rate of disappearance of N_2O_5.

Solution:

$$\text{Rate of reaction} = +\frac{1}{4}\frac{d[NO_2]}{dt} = \frac{1}{4} \times 0.0072 = 0.0018 \text{ mol L}^{-1} \text{ s}^{-1}$$

$$\text{Rate of disappearance of } N_2O_5 = -\frac{d[N_2O_5]}{dt} = 2 \times \text{Rate} = 2 \times 0.0018 = 0.0036 \text{ mol L}^{-1} \text{ s}^{-1}$$

Factors Influencing Rates of Chemical Reactions

1. **Reactant Concentration:** Elevates collision frequency per unit volume, directly scaling rate according to the differential rate law.
2. **Surface Area in Heterogeneous Systems:** Subdividing solid reactants into fine powders exposes more active surface sites, dramatically accelerating reaction velocity.
3. **Temperature:** A 10° C rise typically doubles or triples reaction rates by exponentially increasing the molecular fraction exceeding activation energy E_a.
4. **Catalyst Action:** Provides an alternate mechanism with lowered E_a, accelerating both forward and reverse rates without affecting equilibrium position.
5. **Photochemical Activation:** Light photons ($h\nu$) supply activation energy in specific reactions (e.g., $H_2 + Cl_2 \xrightarrow{h\nu} 2\ HCl$).

Concept Module 2: Rate Law, Order & Molecularity

The Rate Law Formulation

The **Rate Law** is the mathematical expression in which reaction rate is expressed in terms of the molar concentrations of reactants with each term raised to some power:

$$\text{Rate} = k[A]^x[B]^y$$

where k is the **Rate Constant** (or **Specific Reaction Rate**).

- **Critical Distinction:** The exponents x and y are **determined solely by experiment** and may or may not equal the stoichiometric coefficients a and b of the balanced chemical equation.

Order of a Reaction

The **Overall Order (n)** is the sum of the powers of the concentration terms in the experimental rate law:

$$n = x + y$$

- A reaction can have an order of zero, integer $(1, 2, 3)$, fraction $(\frac{1}{2}, \frac{3}{2})$, or even negative.
- **Units of Rate Constant (k):**

$$\text{Units of } k = (\text{mol L}^{-1})^{1-n} \times \text{s}^{-1}$$

  - **Zero Order ($n = 0$):** mol L^{-1} s^{-1} (same as rate units).
  - **First Order ($n = 1$):** s^{-1} or min^{-1} (independent of concentration units).
  - **Second Order ($n = 2$):** L mol^{-1} s^{-1} (or mol^{-1} L s^{-1}).
  - **Third Order ($n = 3$):** L^2 mol^{-2} s^{-1}.

- **Half Order ($n = \frac{1}{2}$):** $\text{mol}^{\frac{1}{2}}\ \text{L}^{-\frac{1}{2}}\ \text{s}^{-1}$.

Determination of Rate Law from Initial Rate Data

In laboratory kinetics, the method of initial rates measures the initial reaction rate over several runs while varying one reactant concentration at a time:

WORKED EXAMPLE 2 — INITIAL RATES METHOD ANALYSIS

Consider the reaction $2A + B \rightarrow C + D$. The initial rate was measured at several different concentrations:

$[A]$ (mol/L)	$[B]$ (mol/L)	Initial Rate (mol/L·s)
0.10	0.10	2.0×10^{-3}
0.20	0.10	8.0×10^{-3}
0.10	0.20	4.0×10^{-3}

Determine: (a) Order with respect to A, (b) Order with respect to B, (c) Rate law, and (d) Rate constant k.

Solution: Let Rate $= k[A]^x[B]^y$.

- Comparing Exp 1 & 2 (with $[B]$ held constant at 0.10):

$$\frac{\text{Rate}_2}{\text{Rate}_1} = \frac{8.0 \times 10^{-3}}{2.0 \times 10^{-3}} = \left(\frac{0.20}{0.10}\right)^x \implies 4 = 2^x \implies x = 2$$

- Comparing Exp 1 & 3 (with $[A]$ held constant at 0.10):

$$\frac{\text{Rate}_3}{\text{Rate}_1} = \frac{4.0 \times 10^{-3}}{2.0 \times 10^{-3}} = \left(\frac{0.20}{0.10}\right)^y \implies 2 = 2^y \implies y = 1$$

- Rate Law: $\mathbf{Rate = k[A]^2[B]}$ (Overall order $= 2 + 1 = 3$).
- Solving for k using Exp 1:

$$k = \frac{\text{Rate}}{[A]^2[B]} = \frac{2.0 \times 10^{-3}}{(0.10)^2 \times 0.10} = \frac{2.0 \times 10^{-3}}{1.0 \times 10^{-3}} = 2.0\ \text{L}^2\ \text{mol}^{-2}\ \text{s}^{-1}$$

Molecularity of an Elementary Reaction

The **Molecularity** is the number of reacting species (atoms, ions, or molecules) taking part in an elementary step which must collide simultaneously to bring about chemical change.

- **Unimolecular (1):** $NH_4NO_2 \rightarrow N_2 + 2H_2O$
- **Bimolecular (2):** $2HI \rightarrow H_2 + I_2$
- **Trimolecular (3):** $2NO + O_2 \rightarrow 2NO_2$ (Reactions with molecularity > 3 are extremely rare because simultaneous collision of 4 or more particles is statistically improbable).

Comparison Parameter	Order of Reaction	Molecularity of Reaction
Definition	Sum of concentration exponents in experimental rate law.	Number of species colliding in an elementary step.

Comparison Parameter	Order of Reaction	Molecularity of Reaction
Nature	Strictly experimental quantity; cannot be deduced from equation.	Theoretical concept deduced from reaction mechanism.
Allowable Values	Can be zero, whole number, fraction, or negative.	Only positive whole integers (1, 2, or 3); never zero or fraction.
Scope	Applicable to both elementary and complex reactions.	Meaningful **only** for elementary single-step reactions.
Dependence	Can change with experimental temperature or pressure.	Invariant; fixed solely by the reaction mechanism.

Complex Multi-Step Reactions & The Rate Determining Step (RDS)

Most chemical transformations occur via consecutive elementary steps (**reaction mechanism**):

- **The Rate Determining Step (RDS):** The slowest elementary step acts as the kinetic bottleneck, dictating the overall reaction velocity.
- **Thermal Decomposition of N_2O_5 ($2N_2O_5 \rightarrow 4NO_2 + O_2$):** Proceeds via: (1) $N_2O_5 \leftrightarrow NO_2 + NO_3$ (fast equilibrium); (2) $NO_2 + NO_3 \rightarrow NO + NO_2 + O_2$ (slow, RDS); (3) $NO + NO_3 \rightarrow 2NO_2$ (fast). Substituting the equilibrium concentration of intermediate gives the experimental first-order rate law: Rate $= k[N_2O_5]$.
- **Anomalous Negative Temperature Coefficient:** Oxidation of nitric oxide ($2NO + O_2 \rightarrow 2NO_2$) slows down upon heating because a rapid exothermic pre-equilibrium ($2NO \leftrightarrow N_2O_2, \Delta H < 0$) shifts backward at higher temperatures faster than the rate constant increases.

Concept Module 3: Integrated Rate Equations & Half-Life

1. Zero Order Reactions

A reaction where the rate is proportional to the zero power of the reactant concentration:

$$-\frac{d[R]}{dt} = k[R]^0 = k \Longrightarrow -d[R] = kdt$$

Integrating from $t = 0$ ($[R] = [R]_0$) to time t ($[R]$):

$$[R] = [R]_0 - kt \quad \text{or} \quad k = \frac{[R]_0 - [R]}{t}$$

- **Linear Plot:** Plot of $[R]$ against t is a straight line with:
  - y-intercept $= [R]_0$
  - Slope $= -k$
- **Half-Life ($t_{(\frac{1}{2})}$):** The time required for reactant concentration to reduce to half its initial value ($[R] = \frac{[R]_0}{2}$):

$$t_{\frac{1}{2}} = \frac{[R]_0}{2k}$$

Key Rule: For zero order, $t_{\frac{1}{2}}$ is **directly proportional** to initial concentration $[R]_0$.

- **Time for 100% Completion:** $t_{100\%} = \frac{[R]_0}{k} = 2 \times t_{\frac{1}{2}}$.

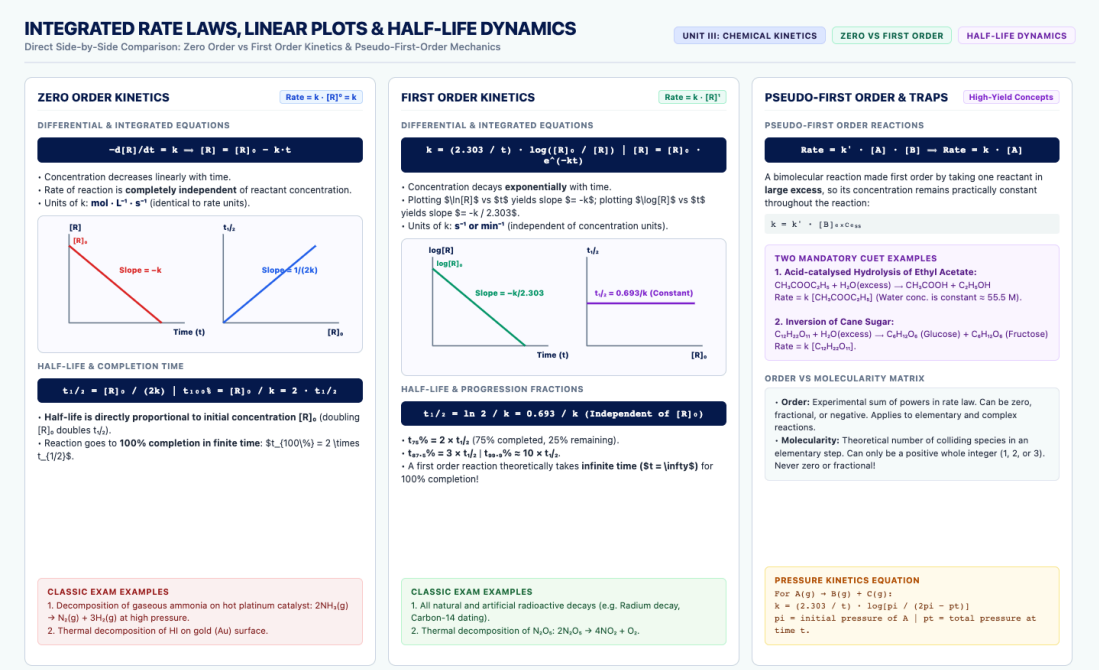


Figure 1: Integrated Rate Laws: Zero Order vs First Order linear kinetics, half-life concentration invariance, and pseudo-first-order solvent excess mechanics.

2. First Order Reactions

A reaction where the rate is directly proportional to the first power of reactant concentration:

$$-\frac{d[R]}{dt} = k[R] \Longrightarrow \frac{d[R]}{[R]} = -kdt$$

Integrating: $\ln[R] = -kt + \ln[R]_0 \Longrightarrow [R] = [R]_0 e^{-kt}$ Converting to common logarithm (base 10):

FIRST ORDER INTEGRATED RATE EQUATIONS

$$k = \frac{2.303}{t} \log\left(\frac{[R]_0}{[R]}\right) = \frac{2.303}{t} \log\left(\frac{a}{a-x}\right)$$

$$\log[R] = \log[R]_0 - \left(\frac{k}{2.303}\right)t$$

- Plotting $\log[R]$ vs t yields a straight line with y-intercept $= \log[R]_0$ and **Slope** $= -\frac{k}{2.303}$.

Half-Life of a First Order Reaction ($t_{(\frac{1}{2})}$):

Substitute $[R] = \frac{[R]_0}{2}$ at $t = t_{\frac{1}{2}}$:

$$t_{\frac{1}{2}} = \frac{2.303}{k} \log\left(\frac{[R]_0}{\frac{[R]_0}{2}}\right) = \frac{2.303}{k} \log 2 = \frac{2.303 \times 0.3010}{k}$$

$$t_{\frac{1}{2}} = \frac{0.693}{k}$$

FUNDAMENTAL FIRST ORDER HALF-LIFE RULES

1. **Concentration Independence:** $t_{\frac{1}{2}}$ is **completely independent** of initial concentration $[R]_0$. Doubling or tripling $[R]_0$ has zero impact on the time required to consume 50% of the reactant.
2. **Progressive Multi-Half-Life Relations:**
   - $t_{75\%} = 2 \times t_{50\%}$ (75% completed $\rightarrow$ 25% remaining $= \left(\frac{1}{2}\right)^2$).
   - $t_{87.5\%} = 3 \times t_{50\%}$ (12.5% remaining $= \left(\frac{1}{2}\right)^3$).
   - $t_{99.9\%} \approx 10 \times t_{50\%}$ ($[R] = \frac{[R]_0}{1000}$).
3. **Radioactive Decay:** All natural and artificial radioactive disintegrations follow **first-order kinetics** ($N = N_0 e^{-\lambda t}, t_{\frac{1}{2}} = \frac{0.693}{\lambda}$).

WORKED EXAMPLE 3 – FIRST ORDER HALF-LIFE PROOFS

Prove that for a first order reaction, the time required for 99.9% completion is approximately 10 times the half-life ($t_{(\frac{1}{2}\}}$).

Solution:

1. For half-life (50% completion):

$$t_{50\%} = \frac{2.303}{k} \log\left(\frac{[R]_0}{0.50[R]_0}\right) = \frac{2.303}{k} \log 2 = \frac{0.693}{k}$$

2. For 99.9% completion ($[R] = [R]_0 - 0.999[R]_0 = 0.001[R]_0$):

$$t_{99.9\%} = \frac{2.303}{k} \log\left(\frac{[R]_0}{0.001[R]_0}\right) = \frac{2.303}{k} \log 10^3 = \frac{2.303 \times 3}{k} = \frac{6.909}{k}$$

3. Comparing the two times:

$$\frac{t_{99.9\%}}{t_{50\%}} = \frac{\frac{6.909}{k}}{\frac{0.693}{k}} = 9.97 \approx 10 \Longrightarrow t_{99.9\%} \approx 10 \times t_{\frac{1}{2}}$$

Comprehensive Kinetics Comparison Across Reaction Orders

Kinetic Feature	Zero Order ($n = 0$)	First Order ($n = 1$)	Second Order ($n = 2$)
Differential Rate	$-\frac{d[R]}{d}t = k$	$-\frac{d[R]}{d}t = k[R]$	$-\frac{d[R]}{d}t = k[R]^2$
Integrated Form	$[R] = [R]_0 - kt$	$\ln\left(\frac{[R]_0}{[R]}\right) = kt$	$\frac{1}{[R]} - \frac{1}{[R]_0} = kt$
Linear Plot	$[R]$ vs t (Slope $= -k$)	$\log[R]$ vs t (Slope $= -\frac{k}{2.303}$)	$\frac{1}{[R]}$ vs t (Slope $= +k$)
Half-Life ($t_{\frac{1}{2}}$)	$t_{\frac{1}{2}} = \frac{[R]_0}{2k}$ ($\propto [R]_0$)	$t_{\frac{1}{2}} = \frac{0.693}{k}$ ($\propto [R]_0^0$)	$t_{\frac{1}{2}} = \frac{1}{k[R]_0}$ ($\propto \frac{1}{[R]_0}$)
100% Completion	$t_{100\%} = \frac{[R]_0}{k} = 2t_{\frac{1}{2}}$	$t_{100\%} = \infty$ (Infinite time)	$t_{100\%} = \infty$ (Infinite time)
Units of k	mol L^{-1} s^{-1}	s^{-1}	L mol^{-1} s^{-1}

Concept Module 4: Gaseous Kinetics & Pseudo-First-Order

Integrated Rate Equation for Gaseous Phase Reactions

For a typical first-order gas-phase decomposition:

$$A(g) \rightarrow B(g) + C(g)$$

- Let initial pressure of A at $t = 0$ be p_i.
- At time t, let x atm of A decompose to form x atm of B and x atm of C:
  - Partial pressure of A: $p_A = p_i - x$
  - Partial pressure of B: $p_B = x$
  - Partial pressure of C: $p_C = x$
- Total pressure at time t (p_t):

$$p_t = p_A + p_B + p_C = (p_i - x) + x + x = p_i + x \Longrightarrow x = p_t - p_i$$

- Substituting x back into p_A:

$$p_A = p_i - (p_t - p_i) = 2p_i - p_t$$

- Applying the first-order rate equation:

$$k = \frac{2.303}{t} \log\left(\frac{p_i}{p_A}\right) = \frac{2.303}{t} \log\left(\frac{p_i}{2p_i - p_t}\right)$$

Pseudo-First-Order Reactions

A reaction that is truly of higher molecularity (e.g. bimolecular) but behaves kinetically as **first order** because one of the reacting species is present in **huge stoichiometric excess**.

- Because the concentration of the excess reactant changes negligibly during the entire reaction, its concentration is absorbed into an effective pseudo-first-order rate constant k:

$$\text{Rate} = k'[A][B]_{\text{excess}} \Longrightarrow \text{Rate} = k[A] \quad (\text{where} \quad k = k'[B]_{\text{excess}})$$

Classic CUET UG Examples of Pseudo-First-Order Reactions

1. **Acid-Catalysed Hydrolysis of Ethyl Acetate:**

$$CH_3COOC_2H_5 + H_2O(\text{excess}) \xrightarrow{H^+} CH_3COOH + C_2H_5OH$$

Water is present at ≈ 55.5 M, while ester concentration is ≈ 0.01 M. Even when all ester reacts, water concentration changes from 55.5 M to 55.49 M, remaining effectively constant.

$$\text{Rate} = k[CH_3COOC_2H_5]$$

2. **Inversion of Cane Sugar (Sucrose):**

$$C_{12}H_{22}O_{11} + H_2O(\text{excess}) \xrightarrow{H^+} C_6H_{12}O_6(\text{D-glucose}) + C_6H_{12}O_6(\text{D-fructose})$$

- **Polarimeter Observation:** Sucrose is dextrorotatory ($+66.5^0$). Upon hydrolysis, equimolar D-glucose ($+52.5^0$) and D-fructose (-92.4^0) form. Because fructose's laevorotation exceeds glucose's dextrorotation, the overall optical rotation flips from dextro ($+$) to laevo ($-$), hence termed **inversion of sugar**.

$$k = \frac{2.303}{t} \log \frac{r_0 - r_\infty}{r_t - r_\infty}$$

where r_0, r_t, and r_∞ are optical angle readings at $t = 0, t = t$, and $t = \infty$.

Concept Module 5: Temperature Dependence & Arrhenius Theory

For almost all chemical reactions, an increase in temperature markedly increases the reaction rate.

- **The Temperature Coefficient:** For every 10^0C rise in temperature, the rate constant doubles or triples:

$$\text{Temperature Coefficient} = \frac{k_{T+10}}{k_T} \approx 2 \text{ to } 3$$

The Arrhenius Equation

Svante Arrhenius proposed the quantitative formulation:

$$k = Ae^{-\frac{E_a}{RT}}$$

- k: Rate constant.
- A: **Arrhenius Pre-Exponential Factor** (Frequency Factor), representing the collision frequency of molecules.
- E_a: **Activation Energy** (in J mol^{-1} or kJ mol^{-1}).
- R: Universal gas constant (8.314 J K^{-1}mol^{-1}).
- T: Absolute temperature in Kelvin.
- $e^{-\frac{E_a}{RT}}$: The **Boltzmann Fraction** of molecules possessing kinetic energy greater than or equal to E_a.

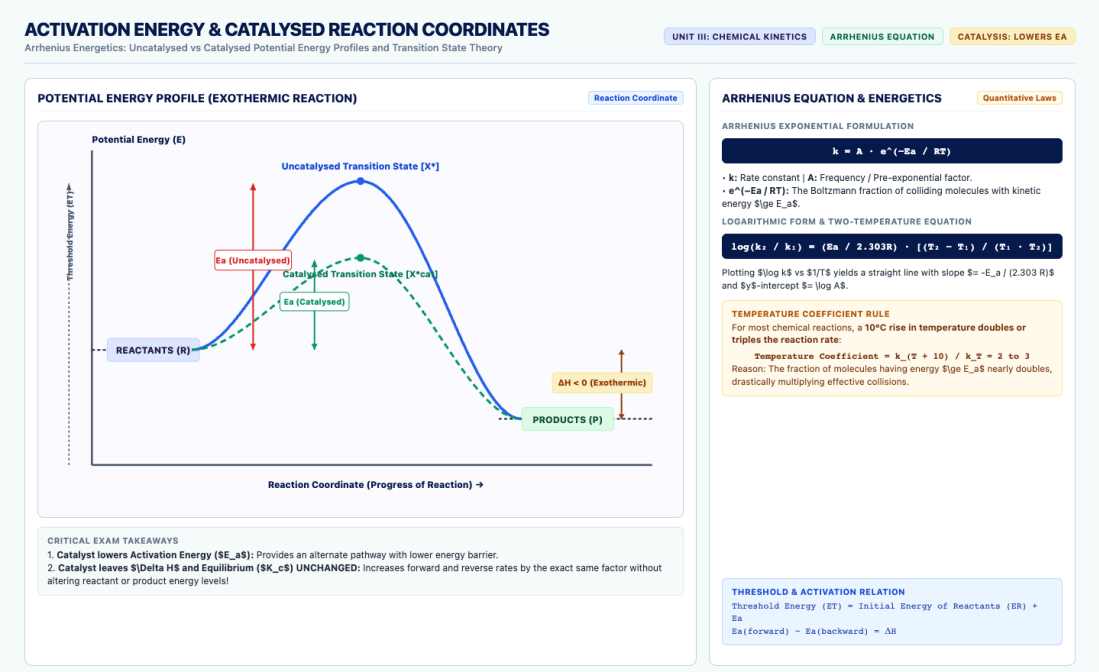


Figure 2: Potential Energy Profile: Activation energy barrier (E_a), transition state (X*), and the lowering of E_a via an alternate catalysed reaction pathway.

Logarithmic Forms of Arrhenius Equation

Taking natural logarithm (ln):

$$\ln k = \ln A - \frac{E_a}{RT}$$

Converting to common base-10 logarithm:

$$\log k = \log A - \frac{E_a}{2.303RT}$$

- **Linear Arrhenius Plot:** Plotting $\log k$ against $\frac{1}{T}$ yields a straight line:
  - y-intercept $= \log A$
  - **Slope** $= -\frac{E_a}{2.303R}$
  - $E_a = -2.303R \times$ Slope

Calculation of E_a at Two Different Temperatures

If rate constants are k_1 and k_2 at absolute temperatures T_1 and T_2:

TWO-TEMPERATURE ARRHENIUS EQUATION

$$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{T_2 - T_1}{T_1 \times T_2}\right]$$

WORKED EXAMPLE 4 — ARRHENIUS ACTIVATION ENERGY CALCULATION

The rate constant of a reaction doubles when the temperature increases from 298 K to 308 K. Calculate the activation energy (E_a) of the reaction. ($R = 8.314$ J K^{-1}mol^{-1}, $\log 2 = 0.3010$).

Solution: Given $\frac{k_2}{k_1} = 2, T_1 = 298$ K, $T_2 = 308$ K.

$$\log 2 = \frac{E_a}{2.303 \times 8.314}\left[\frac{308 - 298}{298 \times 308}\right]$$

$$0.3010 = \frac{E_a}{19.147}\left[\frac{10}{91784}\right] \Longrightarrow E_a = \frac{0.3010 \times 19.147 \times 91784}{10}$$

$$E_a = 52897 \text{ J mol}^{-1} = 52.9 \text{ kJ mol}^{-1}$$

Maxwell-Boltzmann Energy Distribution & The 10 K Rate Doubling Rule

At temperature T, molecules possess a distribution of kinetic energies with peak at the most probable kinetic energy (E_{mp}). When temperature rises to $T + 10$:

1. The curve flattens and broadens, shifting toward higher energies.
2. The fraction of molecules possessing energy $E \geq E_a$ (the shaded area beyond E_a) doubles or triples:

$$x = \frac{N_E}{N_T} = e^{-\frac{E_a}{RT}}$$

Taking logarithms: $\log x = -\frac{E_a}{2.303RT}$, confirming the exponential sensitivity of reaction rate to temperature.

Concept Module 6: Collision Theory & Mechanism of Catalysis

Collision Theory of Chemical Reactions

Developed by Max Trautz and William Lewis, based on the kinetic molecular theory of gases.

- **Fundamental Postulate:** Reactant molecules are hard spheres; reactions occur when molecules collide with one another.
- **Rate of Reaction Formula:**

$$\text{Rate} = P \times Z_{AB} \times e^{-\frac{E_a}{RT}}$$

where Z_{AB} is the **collision frequency** (number of collisions per second per unit volume), and P is the **Steric Factor** (Orientation Factor).

- **Two Criteria for an Effective Collision:**
  1. **Energy Barrier:** Colliding molecules must possess a minimum energy termed the **Threshold Energy (E_T)**:

$$E_T = \text{Initial Energy of Reactants} + E_a$$

  2. **Orientation Barrier:** Reacting molecules must collide with proper spatial orientation so that reactive chemical bonds can break and new bonds can form.

Catalysis & Its Impact on Chemical Energetics

A **catalyst** is a substance that alters the rate of a chemical reaction without undergoing any permanent chemical change itself.

- **Action Mechanism:** An intermediate complex forms between reactants and catalyst, providing an **alternate reaction pathway with a lower activation energy barrier (E_a)**.
- **Key Exam Facts Regarding Catalysts:**
  1. A catalyst **lowers E_a** equally for both forward and reverse reactions.
  2. A catalyst **increases reaction rate** equally in both directions, accelerating the attainment of equilibrium.
  3. A catalyst **DOES NOT alter:**
     - The enthalpy change of reaction ($\Delta_r H$).
     - The Gibbs free energy of reaction ($\Delta_r G$).
     - The equilibrium constant (K_c).
     - The equilibrium concentrations of reactants and products.

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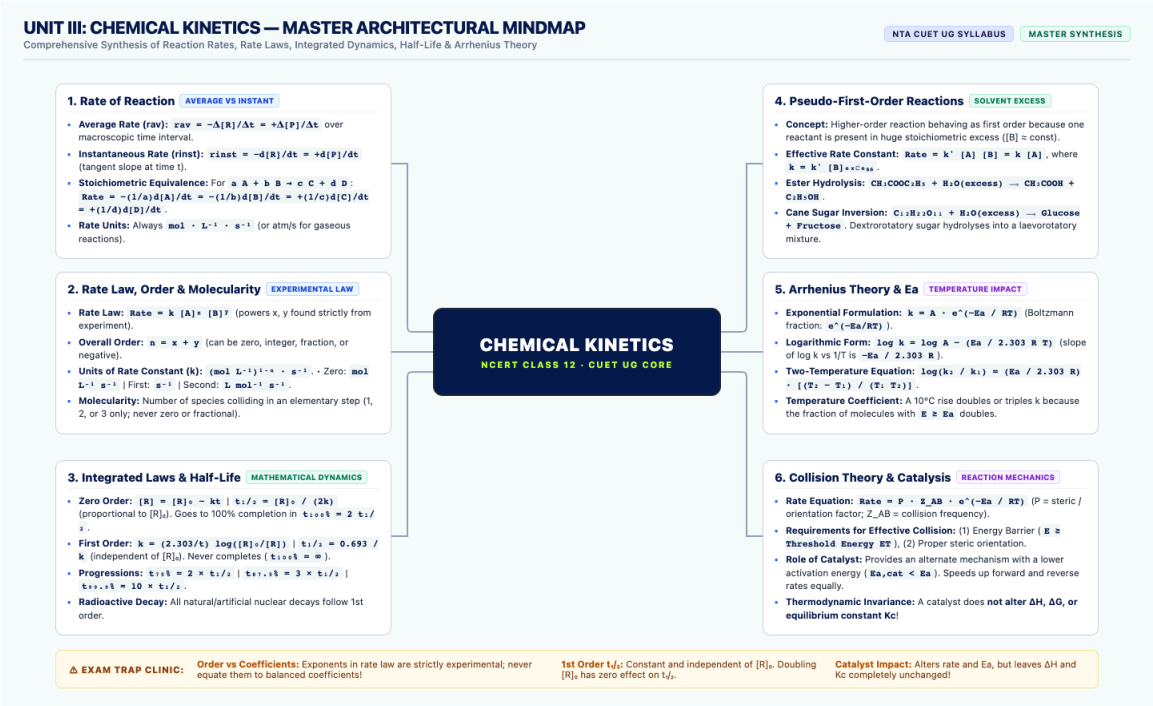


Figure 3: Unit III Chemical Kinetics Master Mindmap: Comprehensive synthesis of rates, order vs molecularity, integrated rate laws, temperature scaling, Arrhenius kinetics, and collision theory.

MASTER KINETICS ARCHITECTURE & REACTION DYNAMICS BREAKDOWN

The cognitive map above consolidates the mathematical foundations and core concepts across Unit III:

- **Rate Law Formulation:** Always differentiate between rate units (mol L^{-1} s^{-1} universally) and rate constant units ($(\text{mol } L^{-1})^{1-n}$ s^{-1}). Order n can be fractional or zero; molecularity is strictly a positive integer $(1, 2, 3)$.
- **Integrated Kinetics & Half-Life Multipliers:** For zero order, $t_{\frac{1}{2}} \propto [R]_0$ and $t_{100\%} = 2 \times t_{\frac{1}{2}}$. For first order, $t_{\frac{1}{2}}$ is concentration-independent ($\frac{0.693}{k}$), yielding fixed milestones: $t_{75\%} = 2t_{50\%}, t_{87.5\%} = 3t_{50\%}$, and $t_{99.9\%} \approx 10t_{50\%}$.
- **Gaseous Reactions Pressure Kinetics:** In gas-phase decompositions $A(g) \rightarrow B(g) + C(g)$, remember that partial pressure $p_A = 2p_i - p_t$. Never substitute raw total pressure directly into the logarithmic ratio!
- **Arrhenius Energetics:** On the $\log k$ vs $\frac{1}{T}$ plot, the slope is strictly $-\frac{E_a}{2.303R}$. Catalysts lower the activation energy equally for both forward and reverse reactions, increasing rate constants without altering $\Delta H, \Delta G$, or equilibrium constant K_{eq}.

UNIT III MASTER KINETICS FORMULA & HIGH-YIELD CHEATSHEET

Core Concept / Law	Mathematical Expression	Application & CUET Problem Strategy
Stoichiometric Rate	$\text{Rate} = -\frac{1}{a}\frac{d[A]}{dt} = +\frac{1}{c}\frac{d[C]}{dt}$	Divide appearance/disappearance rates by stoichiometric coefficients.
Rate Constant Units	$k = (\text{mol L}^{-1})^{1-n}\text{s}^{-1}$	Zero order: mol L^{-1} s^{-1}; 1st: s^{-1}; 2nd: L mol^{-1} s^{-1}.
Zero Order Integrated	$[R] = [R]_0 - kt \mid t_{\frac{1}{2}} = \frac{[R]_0}{2k}$	Linear $[R]$ vs t plot with slope $-k$. Half-life is proportional to $[R]_0$.
First Order Integrated	$k = \left(\frac{2.303}{t}\right)\log\left(\frac{[R]_0}{[R]}\right)$	Linear $\log[R]$ vs t with slope $-\frac{k}{2.303}$. Concentration-independent $t_{\frac{1}{2}}$.
First Order Milestones	$t_{75\%} = 2t_{50\%} \mid t_{99.9\%} \approx 10t_{50\%}$	Every additional half-life halves the remaining reactant quantity.
Gaseous Decomposition	$k = \left(\frac{2.303}{t}\right)\log\left(\frac{p_i}{2p_i - p_t}\right)$	Applies to $A(g) \rightarrow B(g) + C(g)$ where partial pressure $p_A = 2p_i - p_t$.
Pseudo-First-Order	$\text{Rate} = k'[A][H_2O] = k[A]$	Ester hydrolysis and cane sugar inversion ($k = \left(\frac{2.303}{t}\right)\log\left(\frac{r_0 - r_\infty}{r_t - r_\infty}\right)$).
Arrhenius Equation	$k = Ae^{-\frac{E_a}{RT}}$	Fraction of molecules with $E \geq E_a$ is $e^{-\frac{E_a}{RT}}$.
Arrhenius Linear Form	$\log k = \log A - \frac{E_a}{2.303RT}$	Plot $\log k$ vs $\frac{1}{T}$: y-intercept $= \log A$; Slope $= -\frac{E_a}{2.303R}$.
Two-Temperature Ratio	$\log\left(\frac{k_2}{k_1}\right) = \left(\frac{E_a}{2.303R}\right)\left[\frac{T_2 - T_1}{T_1T_2}\right]$	Rate doubles every 10 K rise if $E_a \approx 53\text{kJ mol}^{-1}$.
Collision Theory	$\text{Rate} = PZ_{AB}e^{-\frac{E_a}{RT}}$	Requires both threshold kinetic energy and proper steric orientation P.
Catalyst Action	Lowers E_a to $E_{a'}$ equally for fwd/rev	Accelerates rate; does not change $\Delta H, \Delta G$, or equilibrium constant K_{eq}.

TOP 7 HIGH-FREQUENCY CUET TRAPS & PITFALLS

1. **Rate vs Rate Constant Units:** The units of reaction rate are ALWAYS mol L^{-1} s^{-1} regardless of order. The units of rate constant k vary with order: $(\text{mol L}^{-1})^{1-n}$ s^{-1}. For a first order reaction, k is expressed in s^{-1} and is completely independent of concentration units.
2. **Order vs Molecularity:** Order can be zero, fractional, negative, or an integer and is strictly experimentally determined. Molecularity can ONLY be a positive integer $(1, 2, 3)$ and applies strictly to elementary reactions; complex reactions have NO overall molecularity.
3. **Half-Life Concentration Dependence:** For a first order reaction, $t_{\frac{1}{2}} = \frac{0.693}{k}$ is independent of initial concentration $[R]_0$. For a zero order reaction, $t_{\frac{1}{2}} = \frac{[R]_0}{2k}$ is **directly proportional** to $[R]_0$. If doubling $[R]_0$ doubles the half-life, the reaction order is ZERO, not first!
4. **First Order Multipliers vs 100% Completion:** For first order reactions: $t_{75\%} = 2 \times t_{50\%}$, $t_{87.5\%} = 3 \times t_{50\%}$, and $t_{99.9\%} \approx 10 \times t_{50\%}$. A first order reaction theoretically requires infinite time for 100% completion, whereas a zero order reaction completes fully at $t_{100\%} = 2 \times t_{\frac{1}{2}}$.
5. **Gas-Phase Pressure Kinetics ($A(g) \rightarrow B(g) + C(g)$):** At time t, partial pressure $p_A = 2p_i - p_t$ (NOT $p_i - p_t$). The integrated rate law is $k = \left(\frac{2.303}{t}\right)\log\left(\frac{p_i}{2p_i - p_t}\right)$. Always derive p_A from stoichiometry!
6. **Catalyst Thermodynamics:** A catalyst increases the rates of forward and reverse reactions equally by providing an alternate pathway of lower E_a. It has ZERO effect on thermodynamic parameters: ΔH, ΔG, and equilibrium constant K_{eq} remain completely unchanged.
7. **Arrhenius Plot Slope:** For a plot of $\log k$ versus $\frac{1}{T}$, the slope is $-\frac{E_a}{2.303R}$. For a plot of $\ln k$ versus $\frac{1}{T}$, the slope is $-\frac{E_a}{R}$. Forgetting the factor 2.303 in base-10 logarithmic problems is the most common numerical error in CUET!

Practice Questions (Exam Pattern) — Part 1 (Q1 to Q5)

Q1. For the reaction $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, if the rate of disappearance of H_2 is 6.0×10^{-4} mol L^{-1} s^{-1}, the rate of appearance of NH_3 is:

(A) 4.0×10^{-4} mol L^{-1} s^{-1}
(B) 6.0×10^{-4} mol L^{-1} s^{-1}
(C) 9.0×10^{-4} mol L^{-1} s^{-1}
(D) 2.0×10^{-4} mol L^{-1} s^{-1}

Sol. Q1. Rate $= -\left(\frac{1}{3}\right)\left(\frac{d[H_2]}{d}t\right) = +\left(\frac{1}{2}\right)\left(\frac{d[NH_3]}{d}t\right) \Longrightarrow \frac{d[NH_3]}{d}t = \left(\frac{2}{3}\right)(6.0 \times 10^{-4}) = 4.0 \times 10^{-4}$ mol L^{-1} s^{-1}.

Answer: (A)

Q2. The rate constant for a reaction is experimentally found to be $k = 3.5 \times 10^{-4}$ L mol^{-1} s^{-1}. The overall order of this reaction is:

(A) Zero order
(B) First order
(C) Second order
(D) Third order

Sol. Q2. Units of $k = (\text{mol L}^{-1})^{1-n}$ s^{-1}. For L mol^{-1} s^{-1} $= (\text{mol L}^{-1})^{-1}$ s^{-1}, $1 - n = -1 \Longrightarrow n = 2$ (second order).

Answer: (C)

Q3. For reaction $A + B \rightarrow C$, doubling $[A]$ quadruples the rate, while doubling $[B]$ doubles the rate. The overall rate law and total reaction order are:

(A) Rate $= k[A][B]$, Order $= 2$
(B) Rate $= k[A]^2[B]$, Order $= 3$
(C) Rate $= k[A][B]^2$, Order $= 3$
(D) Rate $= k[A]^2[B]^2$, Order $= 4$

Sol. Q3. Rate $\propto [A]^2$ $(2^2 = 4)$ and Rate $\propto [B]^1$ $(2^1 = 2)$. Thus, Rate $= k[A]^2[B]$ with overall order $n = 2 + 1 = 3$.

Answer: (B)

Q4. Which of the following statements regarding the molecularity and order of a chemical reaction is strictly INCORRECT?

(A) Order can be zero, fractional, or negative, but molecularity is always an integer from 1 to 3
(B) Molecularity is applicable to complex multi-step reactions as a whole
(C) Order is determined experimentally, whereas molecularity is deduced from an elementary step
(D) For elementary reactions, the order is always equal to its molecularity

Sol. Q4. Molecularity has no meaning for a complex reaction as a whole; it is meaningful only for individual elementary single-step reactions.

Answer: (B)

Q5. A zero order reaction $A \rightarrow B$ has rate constant $k = 0.050$ mol L^{-1} s^{-1}. If initial concentration $[A]_0 = 0.50$ M, its half-life $(t_{(\frac{1}{2})})$ is:

(A) 10 s
(B) 5.0 s
(C) 2.5 s
(D) 0.50 s

Sol. Q5. For zero order: $t_{\frac{1}{2}} = \frac{[A]_0}{2k} = \frac{0.50}{2 \times 0.050} = 5.0$ s.

Answer: (B)

Practice Questions (Exam Pattern) — Part 2 (Q6 to Q10)

Q6. The half-life of a first order reaction is 69.3 minutes. The value of its rate constant (k) is:

(A) 1.0×10^{-2} min^{-1}
(B) 1.0×10^{-3} min^{-1}
(C) 6.93×10^{-2} min^{-1}
(D) 0.693 min^{-1}

Sol. Q6. For first order: $k = \frac{0.693}{t_{\frac{1}{2}}} = \frac{0.693}{69.3} = 0.010$ min^{-1} $= 1.0 \times 10^{-2}$ min^{-1}.

Answer: (A)

Q7. For a first order chemical reaction, the time required for 75% of the reactant to decompose ($t_{(75\%)}$) is related to its half-life ($t_{(50\%)}$) by:

(A) $t_{75\%} = 1.5 \times t_{50\%}$
(B) $t_{75\%} = 2.0 \times t_{50\%}$
(C) $t_{75\%} = 3.0 \times t_{50\%}$
(D) $t_{75\%} = 4.0 \times t_{50\%}$

Sol. Q7. 75% completion leaves $25\% = \left(\frac{1}{2}\right)^2$ reactant. Each half-life halves the concentration, so $t_{75\%} = 2 \times t_{50\%}$.

Answer: (B)

Q8. A radioactive sample of Carbon-14 ($t_{\frac{1}{2}} = 5730$ years) undergoes first order decay. What fraction of the original activity remains after $17,190$ years?

(A) $\frac{1}{4}$ (25%)
(B) $\frac{1}{8}$ (12.5%)
(C) $\frac{1}{16}$ (6.25%)
(D) $\frac{1}{32}$ (3.125%)

Sol. Q8. Number of half-lives $n = \frac{17190}{5730} = 3$. Remaining fraction $= \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$ (12.5%).

Answer: (B)

Q9. Acid-catalysed hydrolysis of ethyl acetate ($CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$) is classified as a pseudo-first-order reaction because:

(A) The acid catalyst completely inhibits the backward reaction
(B) Water is present in large stoichiometric excess, so $[H_2O]$ remains constant
(C) Both ethyl acetate and water react in equimolar amounts
(D) The reaction proceeds via a free-radical chain mechanism

Sol. Q9. Water is the solvent (≈ 55.5 M) present in huge excess. Its concentration remains constant, yielding Rate $= k[CH_3COOC_2H_5]$.

Answer: (B)

Q10. During the acid-catalysed hydrolysis of cane sugar (sucrose), the optical rotation of the reaction mixture changes because:

(A) Dextrorotatory sucrose hydrolyses into equimolar D-glucose and D-fructose with net laevorotation
(B) Dextrorotatory sucrose converts exclusively into dextrorotatory glucose
(C) Water breaks the asymmetric carbon atoms into optically inactive fragments
(D) Sucrose undergoes rapid mutarotation into non-reducing polysaccharide

Sol. Q10. Sucrose ($+66.5^0$) produces D-glucose ($+52.5^0$) and D-fructose (-92.4^0). Fructose's laevorotation dominates, inverting net optical rotation from (+) to (−).

Answer: (A)

Practice Questions (Exam Pattern) — Part 3 (Q11 to Q15)

Q11. According to the Arrhenius equation ($\log k = \log A - \frac{E_a}{2.303RT}$), a plot of $\log k$ against $\frac{1}{T}$ yields a straight line whose slope equals:

(A) $-\frac{E_a}{R}$ **(B)** $-\frac{E_a}{2.303R}$
(C) $+\frac{E_a}{2.303R}$ **(D)** $-\frac{E_a}{R}$

Sol. Q11. Comparing $\log k = \left(-\frac{E_a}{2.303}R\right)\left(\frac{1}{T}\right) + \log A$ with $y = mx + c$ gives Slope $= -\frac{E_a}{2.303R}$.

Answer: (B)

Q12. If the rate of a chemical reaction doubles for every 10^0C rise in temperature, how many times will the rate increase when temperature rises from 20^0C to 50^0C?

(A) 3 times **(B)** 6 times
(C) 8 times **(D)** 16 times

Sol. Q12. Temperature increase $\Delta T = 50 - 20 = 30^0$C, corresponding to $n = \frac{30}{10} = 3$ intervals. Rate increases by $2^n = 2^3 = 8$ times.

Answer: (C)

Q13. Which of the following parameters of a chemical reaction is altered by the addition of a positive catalyst at a constant temperature?

(A) Enthalpy of reaction ($\Delta_r H$) **(B)** Gibbs free energy change ($\Delta_r G$)
(C) Equilibrium constant (K_c) **(D)** Activation energy (E_a)

Sol. Q13. A catalyst lowers the activation energy (E_a) by providing an alternate pathway, but leaves thermodynamic functions ($\Delta H, \Delta G, K_c$) unchanged.

Answer: (D)

Q14. According to collision theory, why do all collisions between reactant molecules not lead to product formation?

(A) Collisions only occur between molecules of the same chemical species **(B)** Only collisions with energy *ge* threshold energy and proper steric orientation are effective
(C) The pre-exponential factor is always negative at room temperature **(D)** Molecules must undergo electrolytic ionization prior to contact

Sol. Q14. Effective collisions require overcoming both the energy barrier ($EgeE_T$) and the steric orientation barrier (P).

Answer: (B)

Q15. For a gas-phase first order reaction $A(g) \rightarrow B(g) + C(g)$, if p_i is initial pressure of A and p_t is total pressure at time t, the rate constant k is:

(A) $k = \left(\frac{2.303}{t}\right)\log\left[\frac{p_i}{2p_i - p_t}\right]$ **(B)** $k = \left(\frac{2.303}{t}\right)\log\left[\frac{p_i}{2p_i - p_t}\right]$
(C) $k = \left(\frac{2.303}{t}\right)\log\left[\frac{2p_i - p_t}{p_i}\right]$ **(D)** $k = \left(\frac{2.303}{t}\right)\log\left[\frac{p_i}{2p_i - p_t}\right]$

Sol. Q15. At time t: $p_A = p_i - x$, $p_B = x$, $p_C = x$. Total $p_t = p_i + x \Longrightarrow x = p_t - p_i$. Hence $p_A = p_i - (p_t - p_i) = 2p_i - p_t$. Substituting into 1st order: $k = \left(\frac{2.303}{t}\right)\log\left[\frac{p_i}{2p_i - p_t}\right]$.

Answer: (B)